36118 Applied Natural Language Processing

Master of Data Science and Innovation, UTS

Assessment Task 2B

End-to-End NLP Project — Technical Report

Sydney Liveability Explorer

Subject	36118 Applied Natural Language Processing
Submission	AT2 Part B — Group Technical Report
Due Date	Monday, 4 May 2026, 23:59
Suggested Length	10–15 pages (excluding appendix)
Group	Group [Number], Autumn 2026
GitHub Repository	https://github.com/<org-or-user>/<repo>

Team Members

Last Name	First Name	Student ID	GitHub Username

Contents

LLM-Assisted Progress Log

This log is intended for incremental drafting with LLMs. Add one row per meaningful update so the report evolution is auditable and reproducible.

Date	Author	Section	Status	What Changed	Next Action
2026-04-23	LLM	Template Init	Done	Initialized LaTeX report structure from DOCX template.	Start drafting Section 1 with verified project outcomes.

1 Executive Summary

Write a concise overview of the problem, target users, methods, and key outcomes.

2 Project Objectives and Scope

Define the primary objectives, intended user group, and boundaries of this submission.

3 Project Phases and Team Contributions

Describe project phases and map responsibilities to each team member.

Phase Breakdown Table

Phase	Tasks	Team Member(s)
Phase 1 — Data	Data acquisition and preprocessing from Reddit, PDF, BOCSAR, ArcGIS, OSM, and transport sources.	
Phase 2 — NLP Pipeline	Aspect-level sentiment, TF-IDF extraction, topic modeling, preprocessing pipeline.	
Phase 3 — RAG + LLM	Embeddings, ChromaDB vector store, CrewAI query crew, FastAPI backend.	
Phase 4 — Frontend	Next.js dashboard, Leaflet map, charts, conversational UI, onboarding weights.	
Phase 5 — Evaluation + Report	Traditional vs modern NLP comparison, metrics, report, poster, peer review.	

4 Data Sources

Describe each source, access method, format, granularity, and known limitations.

4.1 Primary Data Sources

Document each source contribution to the pipeline.

4.2 Data Limitations and Ethical Considerations

Discuss data bias, staleness, privacy, consent, and social media ethics.

5 NLP Methods and Techniques

Explain methods, rationale, implementation details, and reproducibility settings.

5.1 Text Preprocessing

Tokenization, stopword removal, lemmatization, and domain-specific cleaning.

5.2 Traditional NLP Approaches

5.2.1 Sentiment Analysis (VADER + TextBlob) — Baseline

Explain baseline polarity and subjectivity workflow.

5.2.2 Topic Modelling (LDA — Gensim)

Document topic count, parameters, labeling strategy, and coherence metrics.

5.2.3 TF-IDF Keyword Extraction

Describe extraction strategy and baseline value.

5.3 Modern NLP Approaches

5.3.1 Sentence Embeddings (all-MiniLM-L6-v2)

Explain chunking strategy and ChromaDB upsert details.

5.3.2 Retrieval-Augmented Generation (RAG + LLM)

Detail query crew orchestration, retrieval filters, and grounded answer generation.

5.3.3 Aspect-Level Sentiment Analysis (DistilRoBERTa)

Describe aspect taxonomy, emotion profiling, and production rationale.

5.4 Traditional vs. Modern NLP Comparison

Compare representative query outputs and discuss trade-offs.

6 Findings and Evaluation

Present core findings and evidence from notebooks and backend outputs.

6.1 NLP Pipeline Results

Summarize aspect scores, emotions, topics, and keyword patterns.

6.2 Model Evaluation

Provide metrics and qualitative error analysis.

6.3 Application Evaluation

Report usability observations from walkthroughs or testing.

6.4 Temporal Trends

Summarize relevant time-based patterns from crime and sentiment data.

7 Outcomes and Value Added

Describe user-facing outcomes and practical value for target groups.

8 Recommendations and Next Steps

List highest-impact extensions and maintenance priorities.

9 Conclusion

Summarize contributions, lessons learned, and future potential.

10 References

Use APA 7th edition for all citations.

A Appendix A — System Architecture Diagram

Insert architecture diagrams and explain key flows.

B Appendix B — Dashboard Screenshots

Insert annotated screenshots of map, charts, chat, onboarding, and debug views.

C Appendix C — Sample Code

Add representative code snippets for ingestion, embeddings, agents, and API endpoints.

D Appendix D — Evaluation Tables and Charts

Include confusion matrices, coherence tables, and charts used in Section 6.

E Appendix E — Traditional vs. Modern NLP Comparison Table

Table 3: Traditional vs. Modern NLP Comparison

Technique	**Type**	**Speed**	**Interpretability**	Role in This Project
TF-IDF Keyword Extraction	**Traditional**	**Fast**	**High**	Baseline retrieval — top terms per suburb
LDA Topic Modelling	**Traditional**	**Medium**	**Medium**	Thematic discovery baseline
VADER + TextBlob	**Traditional**	**Fast**	**High**	Baseline sentiment benchmark
RAG + LLM (CrewAI + ChromaDB)	**Modern**	**Medium**	**Medium**	Grounded, cited conversational answers
DistilRoBERTa Aspect Sentiment	**Modern**	**Medium**	**Medium**	Production sentiment and emotion profiling